



Lab 1 — Build the AI System Inventory for NovaBank

Topic 01: AI Security Governance Foundations and Business Risks

Objective: Analyse an organisation's AI footprint and produce the AI system inventory that every other governance control depends on.

Goal: You are the newly appointed AI Governance Lead at NovaBank, a mid-sized Singapore retail bank. Nobody can tell you how many AI systems the bank runs. You will work from the supplied discovery pack — an IT asset extract, a procurement export and a staff survey — to identify every AI system, classify it, and produce a defensible AI system inventory with named owners and risk tiers.

What you'll build: A completed AI System Inventory (ai-inventory.csv) covering 12 AI systems with owner, purpose, data classes, model, autonomy level and risk tier, plus a shadow-AI findings note.

Tools and data: Mock data pack (discovery-it-assets.csv, discovery-procurement.csv, discovery-staff-survey.csv), spreadsheet tool

Data files in this lab

- data/discovery-it-assets.csv
- data/discovery-procurement.csv
- data/discovery-staff-survey.csv

Step-by-step

1. Open the three discovery files in labs/lab-01-ai-inventory/data/ and read the column headings before you filter anything. Note that no single file contains the whole picture.
2. Identify AI systems in discovery-it-assets.csv. Flag any row whose vendor, product name or description indicates a model, assistant, copilot, chatbot, scoring engine or agent.
3. Cross-check discovery-procurement.csv for AI services bought outside IT — look for line items charged to a business cost centre rather than to IT.
4. Read discovery-staff-survey.csv for shadow AI: tools staff use that appear in neither of the other two files. These are your highest-priority findings.
5. Merge into one register. For each system record: System ID, Name, Business owner, Purpose, Data classes touched, Model/vendor, Hosting, Autonomy level, Personal data (Y/N).

6. Assign an autonomy level to each system using the four tiers: Tier 1 supervised (suggests only), Tier 2 constrained (acts within a fixed allowlist), Tier 3 broad with monitoring, Tier 4 full autonomy.
7. Assign a risk tier (High / Medium / Low) using two questions: does it make or materially influence a decision about a person, and does it touch personal or confidential data?
8. Name an accountable owner for every system. Any system without a named human owner is itself a finding — record it as such.
9. Write a half-page shadow-AI findings note: how many unapproved tools you found, what data they touch, and the single control you would introduce first.

Test it

Your inventory has a row for every AI system in all three files, including at least three shadow-AI tools that appear only in the staff survey. Every row has a named owner, an autonomy tier and a risk tier, and you can justify each High rating in one sentence.

Training use only. The organisation, data and individuals in this lab are fictional. Security techniques taught here must only be applied to systems you own or have written authorisation to test.